Indicative solution of the Project Management case (EN)

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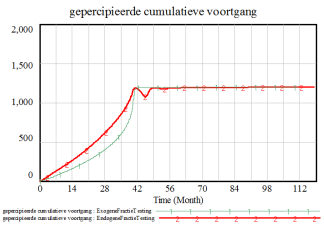
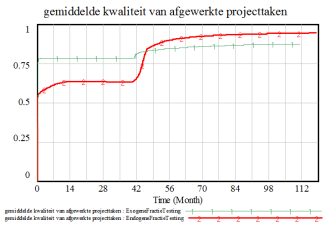
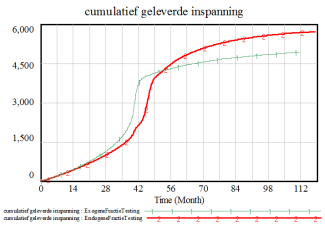
This staged project management case is largely based on George Richardson's Project models[1]. In the first stage, students need to make and simulate a project model without testing and rework in which quality issues remain unnoticed (see Figure 2). In the second stage, students need to extend the first model with undiscovered rework, testing, rework, et cetera. Students then need to validate the extended model, make graphs of the dynamics (see Figures 1(a)-(c)+(d)), interpret the results, add an endogenous testing policy (see Figures 1(a)-(c)+(e)), test it, test the sensitivity of the outcomes for changes in two assumptions, and propose and test a strategy for managing extreme workforce fluctuations. Finally students need to develop an aggregated CLD of the model, as in Figure 3.



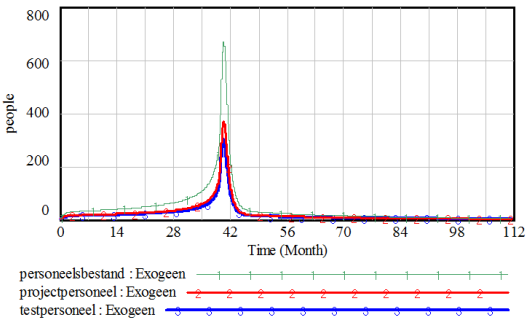
(a) Perceived cumulative progress (endogenous = red)



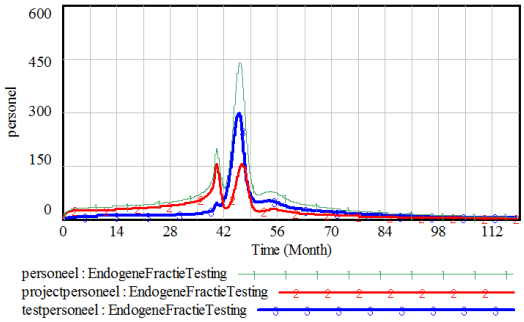
(b) Average quality completed tasks (endogenous = red)



(c) Cumulative effort (endogenous = red)



(d) Personnel (exogenous strategy)



(e) Personnel (endogenous strategy)

Figure 1: Behavior of the model without and with endogenous testing strategy

[1] Available at `http://www.albany.edu/faculty/gpr/PAD624/project1v3.mdl`.

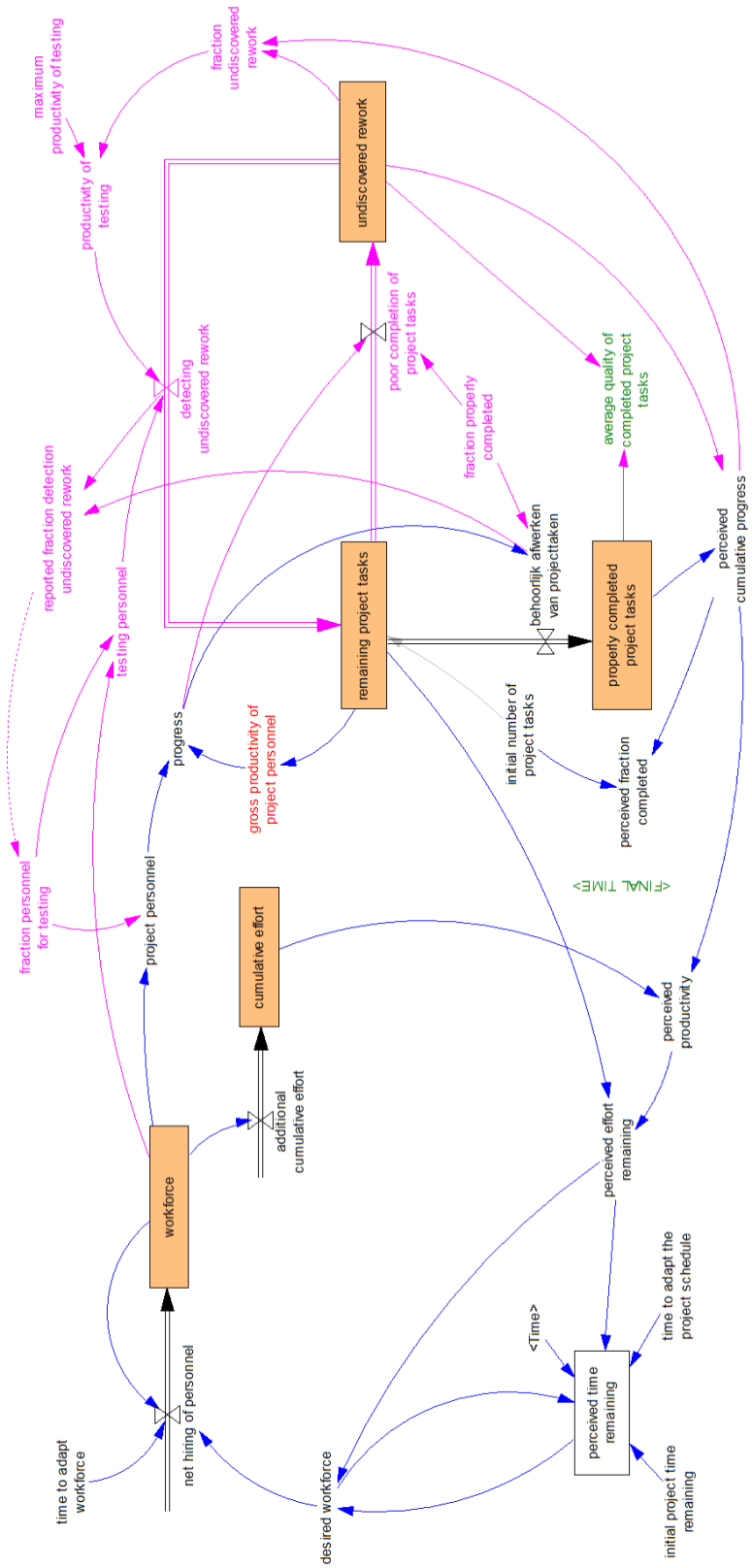
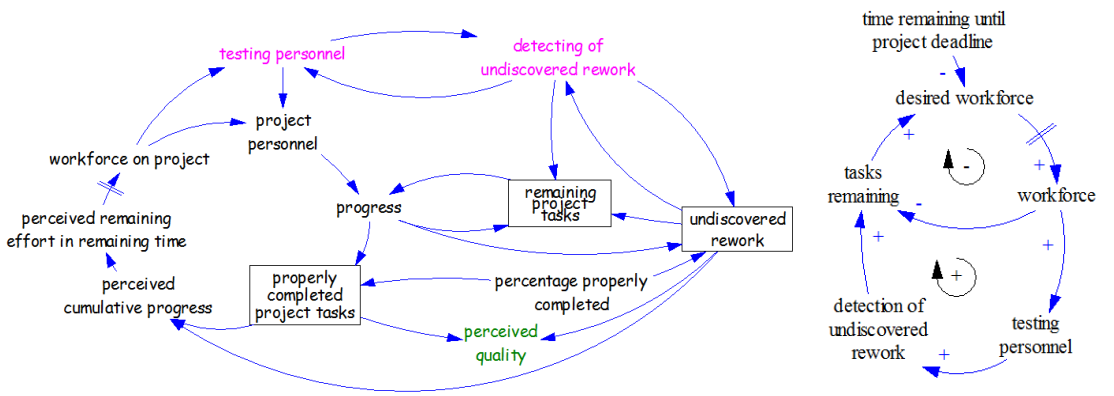


Figure 2: Stock-Flow diagram of the full Project Management model (based on Richardson's project model)



(a) Aggregated CLD by one of the students

(b) Strongly aggregated CLD by one of the students

Figure 3: Aggregated CLDs